ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

L A B R E C O R D

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: R. Hasini

Roll No: _____

Date: _____

Subject Name: Professional English Lab Integrated

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

BIOGRAPHY OF BILL GATES

Bill Gates, whose full name is William Henry Gates III, is a famous American businessman, computer programmer, entrepreneur, and philanthropist. He is best known as the co-founder of Microsoft, one of the world's most successful technology companies. His contribution to computer software and technology has made him one of the most influential people in the technology industry.

Bill Gates was born on October 28, 1955, in Seattle, Washington, in the United States. He grew up in a well-educated family. His father, William H. Gates Sr., was a lawyer, and his mother, Mary Maxwell Gates, was involved in education and community service. Bill had two sisters. From a young age, he showed a strong interest in reading, mathematics, science, and computers.

When Gates was a teenager, he got an opportunity to use a computer at his school. He became very interested in computer programming and spent many hours learning how computers worked and writing programs. During this time, he became friends with Paul Allen, who also had a strong interest in computers. Their friendship later led to the creation of Microsoft.

EDUCATION AND QUALIFICATION:

Bill Gates attended Lakeside School in Seattle, where he developed his interest in computers and programming. After completing school, he joined Harvard University in 1973. He studied subjects including mathematics, computer science, and economics. However, he left Harvard before completing his degree because he decided to concentrate on developing Microsoft with Paul Allen.

Although Gates did not complete a university degree, he gained extensive knowledge and practical experience in computer programming, software development, business, and technology. His success demonstrates that formal qualifications are not the only way to develop expertise. His continuous learning and practical experience played an important role in his career.

SKILLS OF BILL GATES:

Bill Gates has developed many skills throughout his career. One of his most important skills is computer programming. As a young person, he learned programming and used his knowledge to create software. His technical understanding helped him recognize the growing importance of personal computers.

Another important skill is problem-solving. Gates has always been interested in finding solutions to difficult problems. He used this ability in developing software and later in his philanthropic work, where he focused on problems such as disease, poverty, education, and climate change.

Gates also has strong business and entrepreneurial skills. He helped establish Microsoft at a young age and played an important role in its growth. He understood the importance of developing useful software and responding to changes in the technology industry.

He is also known for his leadership and decision-making skills. As Microsoft grew, Gates had to make important decisions about products, business strategies, and the future direction of the company. His leadership helped Microsoft become a major technology company.

Another important skill is communication. Gates has written books, given speeches, participated in interviews, and communicated with business leaders, scientists, governments, and organizations around the world. These activities require the ability to explain complex ideas clearly.

Gates is also known for his research and learning skills. He has a strong interest in reading and learning about science, technology, health, economics, and other subjects. He believes that continuous learning is important for personal and professional development.

CAREER AND MICROSOFT:

In 1975, Bill Gates and Paul Allen founded Microsoft. Their goal

was to create software for personal computers. One of their early successes was developing BASIC software for the Altair computer.

Microsoft continued to grow as personal computers became more popular. The company developed many software products, including the Windows operating system. Windows became widely used in homes, schools, and businesses around the world.

Gates played an important role in Microsoft's development through his knowledge of software, business strategy, leadership, and technology. The success of Microsoft made Gates one of the world's best-known technology entrepreneurs.

PHILANTHROPIC WORK

After becoming successful in the technology industry, Gates devoted a large amount of his time to philanthropy. He and his former wife, Melinda French Gates, established the Bill & Melinda Gates Foundation.

The foundation works on issues such as global health, education, poverty reduction, and development. Gates has supported efforts to prevent diseases, improve healthcare, develop vaccines, and increase educational opportunities.

He has also become interested in climate change and clean energy. Through his investments and charitable activities, he has supported scientific research and technological solutions to environmental problems.

CONCLUSION:

Bill Gates is an excellent example of a person who combined technical knowledge with business skills and entrepreneurship. His qualifications include his education at Lakeside School and Harvard University, while his practical experience in programming and software development became a major part of his expertise.

His most important skills include computer programming, problem-solving, leadership, business management, entrepreneurship, communication, decision-making, research, and

continuous learning. These skills helped him build Microsoft and later contribute to philanthropic activities around the world.

Bill Gates's journey from a young student interested in computers to a successful entrepreneur and philanthropist is an inspiring story. His life shows the importance of curiosity, hard work, innovation, and learning. His achievements in technology and his efforts to address global problems have made him a significant figure in modern history.

MY PROFILE

Name:**R.Hasini**

Email:**hasinisharli@gmail.com**

Linkedin : linkedin.com/in/hasini-rajesh-a2ba0343a

Github:https://github.com/hasini2905

Aspiring GENAI Engineer

EDUCATION:

St.Joseph's College of Engineering

B.E.Computer Science and Engineering

I am a hardworking and quick-learning student with a strong interest in technology and software development. I am passionate about learning new concepts, improving my technical abilities, and exploring the rapidly developing field of computer science.

AREAS OF INTEREST:

- Another major area of interest is software development. I am interested in understanding how software applications are designed, developed, tested, and improved. I would like to learn how different technologies work together to create useful and efficient applications.

- Cloud computing is another important area that interests me. Modern software applications increasingly depend on cloud platforms for computing, storage, databases, networking, and deployment. I would like to understand how applications are deployed and managed in the cloud and learn about technologies used to build scalable and reliable software systems.

- As a CSE student, I understand that strong technical fundamentals are important for building a successful career

in the software industry. Therefore, I aim to develop a strong foundation in several programming languages and core computer science subjects.

GOAL:

- My first goal is to become proficient in C, C++, Python, and Java. Learning multiple programming languages will help me understand different programming concepts and approaches

· I also aim to develop strong knowledge of Data Structures and Algorithms (DSA). I consider DSA an essential part of computer science because it helps in solving problems efficiently. I want to learn different data structures, understand algorithms, analyze their efficiency, and apply them to programming problems. Developing strong algorithmic thinking will also help me become a better software developer.

· Another important area I want to learn is Database Management. Almost every modern application needs to store and manage information effectively. I would like to understand databases, SQL, data organization, database design, and how applications communicate with databases. This knowledge will be particularly useful when developing web and application-based projects.

· In addition to these technical skills, I want to develop a strong understanding of fundamental computer science subjects such as Operating Systems, Computer Networks, Object-Oriented Programming, and Software Engineering. I believe that understanding these concepts will help me become a well-rounded software professional.

MY CAREER GOAL FOR 2030:

· My long-term goal is to build a successful career in the software industry. By the time I graduate in 2030, I aspire to

work as a Software Engineer with expertise in Artificial Intelligence and modern software technologies.

· I would like to develop the skills required to build modern software applications that can solve real-world problems. I also want to understand how AI can be integrated with web applications, cloud platforms, and other software systems.

· To achieve this goal, I plan to continuously improve my programming skills, practice Data Structures and Algorithms, learn database management, explore web and application development, and gain knowledge of cloud computing and artificial intelligence. I also hope to work on practical projects and gain industry experience through internships during my college years.



MY PERSONAL SWOC ANALYSIS

INTRODUCTION:

SWOT Analysis is a useful method for understanding ourselves and planning for the future. SWOT stands for Strengths, Weaknesses, Opportunities, and Threats. We need SWOT analysis to identify our positive qualities, recognize areas that need improvement, understand the opportunities available to us, and prepare ourselves to face possible challenges. It helps us make better decisions, set realistic goals, and develop a clear plan for personal growth.

STRENGTHS:

I consider myself a hardworking, responsible, and positiveminded person. I am interested in learning new things and improving my knowledge. I have good problem-solving, logicalthinking,
and analytical skills. I can adapt to new situations and work well with others. I am also a good listener and respect the ideas and opinions of others. I am determined to achieve my goals and learn from my mistakes. My willingness to learn, patience, and ability to handle challenges are some of my important strengths.

WEAKNESS:

One of my main weaknesses is that I sometimes lack confidence while speaking in front of others. I may feel nervous during presentations or when expressing my ideas to a large group. I also need to improve my communication, timemanagement,
and leadership skills. Sometimes, I overthink situations and take more time to make decisions. I am working on these weaknesses through regular practice, better planning, and developing greater self-confidence.

OPPORTUNITIES:

I have many opportunities to improve myself and build a better future. Online courses, workshops, seminars, internships, competitions, and practical projects can help me gain new knowledge and experience. Participating in group discussions and presentations can improve my communication skills. I can also learn new technologies and stay updated with current trends. Guidance from teachers, mentors, friends, and experienced people can help me identify my strengths and explore new opportunities.

THREATS:

Some challenges that may affect my growth include competition, lack of experience, changing technology, distractions, and poor time management. Fear of failure or lack of confidence may sometimes prevent me from taking new opportunities. To overcome these challenges, I need to stay focused on my goals, continuously improve my skills, manage my time effectively, and maintain a positive attitude.

VISION:

My vision is to become a confident, knowledgeable, independent, and successful person. I want to continuously improve my skills, personality, communication, and knowledge. I aim to face challenges with courage and use every experience as an opportunity to learn. Through hard work, discipline, continuous learning, and determination, I want to achieve my goals and make my family proud.

CONCLUSION:

SWOT analysis helps me understand where I am now and where I want to go in the future. By identifying my strengths, I can use them effectively; by recognizing my weaknesses, I can work to improve them. Understanding opportunities helps me make use of them, while recognizing threats helps me prepare for challenges. Therefore, SWOT analysis is an important tool for self-awareness, personal development, goal setting, and future planning.

My Experience with Write & Improve

I used the Write & Improve website by Cambridge to practise and improve my English writing skills. The activity was interesting because I could choose a topic, write my answer, and submit it to get feedback. The website gave me a level for my writing and showed me the areas where I could improve. It also helped me understand my mistakes in grammar, spelling, vocabulary, and sentence formation.

While doing this activity, I felt that writing in English became easier and more comfortable for me. I was able to check my writing and make changes based on the feedback.. Rewriting my answer after receiving feedback helped me notice my mistakes and improve them.

Overall, I enjoyed using Write & Improve because I could practise on my own and immediately see how I was performing. It helped me improve my confidence in English writing and taught me to check and correct my own work. I learned that regular practice and learning from mistakes can gradually improve my writing skills.



My Experience with Speak & Improve

I enjoyed doing this activity on Speak & Improve because it gave me a chance to practise my English speaking skills in an interesting way. I had to answer different questions by speaking, and after completing the exercises, I received a grade based on my speaking ability. In some exercises, I got a B1 grade, while in another exercise, I got a B2 grade. This helped me understand my current level and also motivated me to improve further.

One thing I found useful was that I could listen to my recorded answers again. By listening to my own voice, I could notice some mistakes in my pronunciation and the way I spoke. I learned that speaking clearly and pronouncing words correctly is important for effective communication. I also became more comfortable speaking in English without taking too much time to think.

Overall, I felt that this activity was both useful and enjoyable. It helped me improve my pronunciation, speaking confidence and ability to express my thoughts in English. The B1 and B2 grades also helped me understand that I can improve my speaking skills with regular practice. I would like to continue practising so that I can become more fluent and confident while speaking English.



NPTEL– Enhancing Soft Skills and Personality

IIT Kanpur | 8-Week Course | Individual Week-wise Summary

Course Overview

The NPTEL course Enhancing Soft Skills and Personality, aims to develop essential soft skills that are important for personal growth, academic success and professional development. The course explains how communication, personality, attitude, emotional intelligence and interpersonal skills influence our everyday lives. It also highlights the importance of maintaining good relationships, managing time, developing positive thinking and behaving professionally. Through this course, I gained useful knowledge that can help me improve myself as a student and prepare for my future career.

Individual Week-wise Summary

Week 1 – Introduction and Growth Mindset

The first week introduces the concept of soft skills and explains their importance in personal and professional life. Soft skills help individuals communicate effectively, work with others and handle different situations confidently. This week also discusses the difference between a fixed mindset and a growth mindset. A fixed mindset limits a person's belief in their abilities, whereas a growth mindset encourages continuous learning and improvement. I learned that failures and mistakes should be considered opportunities to learn rather than reasons to give up. Developing a growth mindset helps us face challenges, build confidence and achieve our goals through consistent effort.

Week 2 – Time Management and Assertiveness

The second week focuses on time management and assertiveness. Time management is an important skill that helps us organise our daily activities and complete tasks within the given time. It explains the importance of setting priorities, planning activities and avoiding unnecessary delays. I understood that procrastination can affect productivity and create stress. This

week also introduces assertiveness, which means expressing our thoughts, feelings and opinions clearly and respectfully. I learned that we should be confident enough to express our ideas while also respecting the opinions of others. These skills are especially useful for students in managing academic responsibilities.

Week 3 – Managing Emotions and Positive Thinking

The third week discusses emotional management and the importance of developing a positive attitude. Every individual experiences different emotions, and learning to control them is essential for maintaining healthy relationships. This week explains the importance of controlling anger, avoiding negative thoughts and responding calmly to difficult situations. Positive thinking helps us approach problems with confidence and find possible solutions instead of becoming discouraged. I learned that our thoughts and attitudes have a strong influence on our behaviour and decisions. By developing emotional control and a positive mindset, we can improve our personal relationships and face challenges more effectively.

Week 4 – People Skills and Personality Development

The fourth week focuses on people skills and the development of a good personality. People skills are essential for interacting with others, building relationships and working successfully in a team. The lessons highlight the importance of kindness, respect, cooperation, patience and understanding different perspectives. It also explains how our behaviour, attitude and communication style influence the impression we create on others. I learned that a good personality is not only about appearance but also about our thoughts, values and actions. Developing interpersonal skills helps us become better team members and build meaningful relationships in both academic and professional environments.

Week 5 – English Language Skills

The fifth week mainly focuses on improving English language skills and understanding common grammatical mistakes. English is an important language for education, communication and

professional opportunities. This week helps learners understand the correct usage of grammar, vocabulary, sentence structures and expressions. It also highlights the importance of avoiding common errors while speaking and writing in English. I learned that effective communication requires not only knowledge of words but also the ability to use them correctly. Practising English regularly can improve fluency, confidence and clarity. These skills will be helpful during presentations, group discussions, interviews and other professional activities.

Week 6 – Humour, Money and Personality

The sixth week discusses the role of humour in personality development and the importance of understanding money. Humour is a useful communication skill that can make conversations more enjoyable and create a friendly atmosphere. Appropriate humour helps reduce tension and improve relationships with others. However, it should always be used carefully without hurting anyone's feelings. The lessons also discuss the relationship between money and personality, highlighting the importance of financial awareness and responsible spending habits. I learned that money should be managed wisely and that we should understand the difference between needs and wants. Maintaining a balanced attitude towards money is important for a responsible lifestyle.

Week 7 – Health and Love in Personality Development

The seventh week highlights the importance of physical health, mental well-being and healthy relationships in developing a balanced personality. Good health is essential for maintaining energy, concentration and productivity. The lessons discuss the importance of regular exercise, nutritious food, proper sleep and healthy daily habits. They also explain the role of love, care, understanding and emotional support in building strong relationships. I learned that personal development is not limited to academic or professional achievements. Taking care of our health and maintaining positive relationships are equally important for a happy and successful life. A healthy mind and body help us

perform better in our daily activities.

Week 8 – Ethics, Etiquette, Memory and Environment

The eighth week focuses on ethics, etiquette, memory improvement and environmental responsibility. Ethics refers to the moral principles that guide our decisions and actions. Professional etiquette teaches us how to behave respectfully in different social and workplace situations. This week also discusses the importance of memory and techniques that can help improve our ability to remember and recall information. In addition, it highlights the responsibility of every individual towards protecting the environment. I learned that honesty, discipline, respect and responsibility are important qualities for becoming a good person. We should also develop environmentally friendly habits and contribute towards creating a cleaner and healthier society.

Overall Course Conclusion

The eight-week NPTEL course Enhancing Soft Skills and Personality provided me with valuable knowledge about personal development and professional behaviour. Each week introduced different skills that are important for becoming a confident and responsible individual. I learned about the importance of communication, time management, positive thinking, emotional control, teamwork, English language skills, financial awareness, health, ethics and professional etiquette.

As a first-year engineering student, I understood that technical knowledge alone is not sufficient to achieve success in a professional career. We must also develop communication skills, problem-solving abilities, confidence and a positive attitude. This course encouraged me to identify areas where I can improve myself and motivated me to practise these skills in my everyday life. I believe the knowledge gained from this course will help me in my academic journey, future career and personal growth.

PROJECT PROPOSAL: Offline Disaster Communication System

1. Executive Summary

ResQMesh is an offline disaster communication system designed to help people communicate with one another during emergencies, even when internet services are unavailable. During natural disasters such as floods, cyclones, earthquakes, and landslides, communication networks may fail due to power cuts, damaged mobile towers, or internet service interruptions. In these situations, people may find it difficult to contact their family members, emergency responders, and nearby people for help.

The main objective of ResQMesh is to provide an alternative way to communicate without depending on the internet or continuous electricity supply. The proposed system uses mesh networking technology, in which nearby devices can communicate directly and relay messages to other devices. Depending on the implementation, technologies such as Bluetooth, Wi-Fi Direct, or dedicated low-power radio modules can be used to establish communication.

ResQMesh aims to make emergency communication more accessible, reliable, and useful when conventional communication systems are disrupted. It can help people share emergency messages, request assistance, and exchange important information during disasters.

2. Problem Statement

Communication is one of the biggest challenges during natural disasters and other emergency situations. Most people depend on mobile networks, internet services, and electricity to communicate. However, these facilities may become unavailable during severe disasters because of damaged infrastructure, power failures, and network congestion.

For example, during a flood, a person trapped inside a building may need to inform nearby people about their location and

request immediate assistance. If mobile networks and internet services are unavailable, contacting others becomes difficult. Similarly, rescue teams may face challenges in receiving information about people who need help.

Existing communication methods may not always be accessible when the usual infrastructure fails. Therefore, there is a need for an alternative communication system that can work without internet connectivity and reduce dependence on conventional networks.

ResQMesh proposes a solution that enables nearby devices to exchange emergency messages directly and forward them through connected devices when possible. The project focuses on improving communication during disasters and helping people stay connected when traditional services are disrupted.

3. How ResQMesh Works

ResQMesh uses the concept of mesh networking, where multiple devices form a network and communicate with one another without requiring a central internet connection.

The system works through the following steps:

Device connection: Nearby phones or compatible devices discover one another using supported wireless communication technologies such as Bluetooth or Wi-Fi Direct.

Sending emergency messages: A user can send a text message, emergency alert, or request for help through the ResQMesh application.

Message forwarding: If the intended receiver is not within direct communication range, an intermediate connected device can relay the message toward the destination, provided a suitable route is available.

Emergency identification: Users may mark messages as urgent and share useful details, such as their location when supported and permitted by the device.

Receiving messages: The receiving device displays the emergency message so that nearby people can respond or pass the information to others.

Low-power operation: The system can be designed to reduce battery consumption. However, each participating device still requires some available power to operate.

For the initial prototype, ResQMesh can focus on offline text messaging between nearby devices. Later versions may include multi-device message forwarding, emergency broadcasts, location sharing, and dedicated low-power communication hardware.

4. Advantages of ResQMesh

No internet required: Users can exchange messages without mobile data or an active internet connection.

Reduced dependence on mobile networks: The system can continue communicating locally when conventional networks are unavailable.

Device-to-device communication: Nearby compatible devices can communicate directly.

Extended communication through relaying: Messages may travel beyond the range of a single connection when intermediate devices are available.

Emergency alerts: Users can send urgent messages to request assistance.

Flexible implementation: The system can begin as a simple mobile application and later be improved with additional hardware and features.

ResQMesh does not automatically guarantee communication over long distances. Its performance depends on the wireless technology, device range, number of participating devices, battery availability, and network conditions.

5. Benefits of ResQMesh

ResQMesh can provide several benefits to individuals and communities during emergencies.

For the public: People may be able to contact nearby individuals and request help even when internet services are unavailable.

For disaster-affected communities: Residents may exchange important information, such as safe routes, local hazards, and requests for essential supplies.

For rescue and relief teams: Emergency messages from affected areas may help responders understand local needs and identify places that require attention, when the system is accessible to them.

For rural and remote areas: The technology could provide an additional communication option in locations where conventional connectivity is limited, provided compatible devices are within communication range.

For future development: ResQMesh can be expanded into a community-based emergency communication platform with features such as priority alerts, message acknowledgements, and low-power radio support.

The system should also consider message privacy, authentication, false emergency alerts, and battery limitations to ensure that it can be used responsibly.

6. Conclusion

ResQMesh is a proposed offline disaster communication system that aims to address communication difficulties during emergencies. By using mesh networking and direct device-to-device communication, it seeks to provide an alternative to internet-dependent communication methods.

The project has the potential to help people exchange emergency messages, request assistance, and share important information when traditional communication infrastructure is disrupted. Its practical effectiveness will depend on the wireless technology

used, device compatibility, available power, and the presence of other participating devices.

In conclusion, ResQMesh aims to make emergency communication more accessible and resilient. With further development and testing, it could become a useful supporting tool for disaster preparedness and community safety.



GITHUB README

Hi,I'm Hasini Rajesh! COMPUTER SCIENCE AND ENGINEERING STUDENT

Welcome to my GitHub profile!

I'm a first-year Computer Science and Engineering Student passionate about technologies,programming and continuous learning. I enjoy exploring new concepts , developing my technical skills and improving my problem solving abilities.

About Me

-Learning and practising C programming.

-Interested in software development and problem-solving.

-Continuously improving my logical thinking ,technical abilities and communication skills.

-Focused on building a strong foundation in computer science.

Currently Learning

- C programming
- Programming fundamentals
- Software Developmnt Basics

Goals

- Develop strong programming fundamentals.
- Build practical and meaningful projects.
- Improve problem-solving skills.
- Gain experience through collaborative projects.
- Grow as a professional software engineer.

connect with me

GitHub:hasini2905

Contact

hasinisharli@gmail.com

www.linkedin.com/in/hasini-rajesh-a2ba0343a (LinkedIn)

Hasini Rajesh

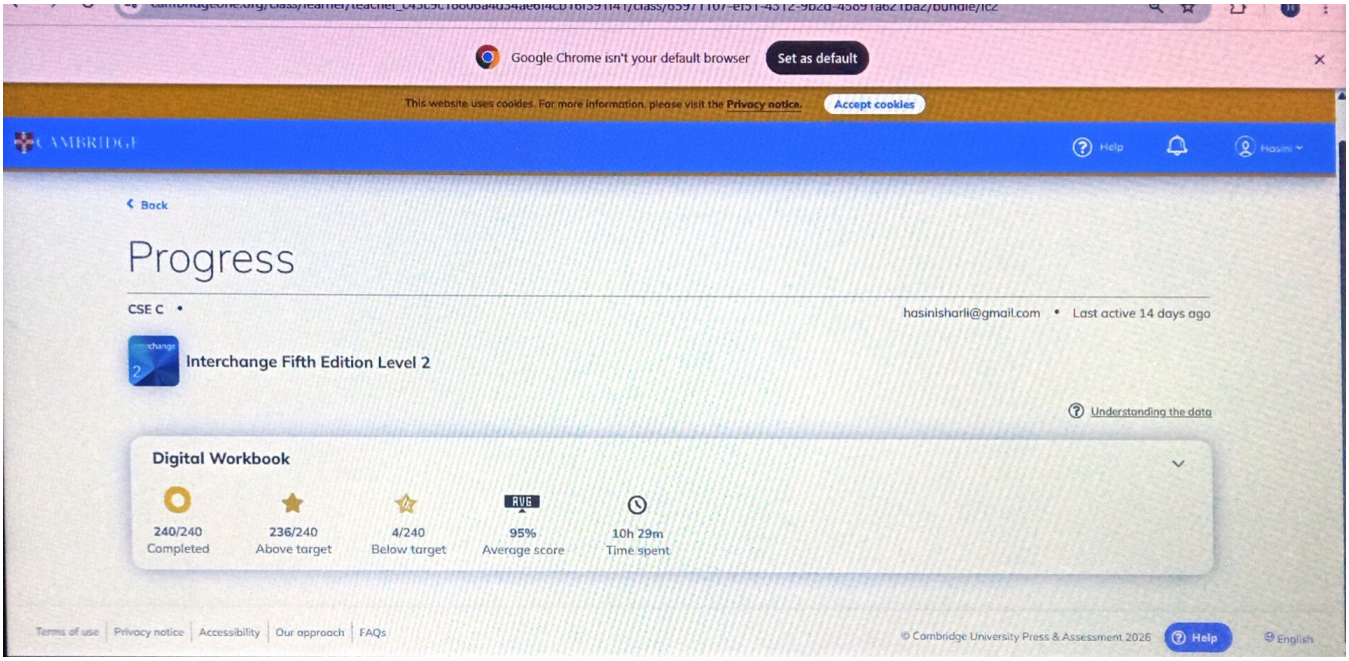
Student at St.Joseph’s college of Engineering
Chennai, Tamil Nadu, India

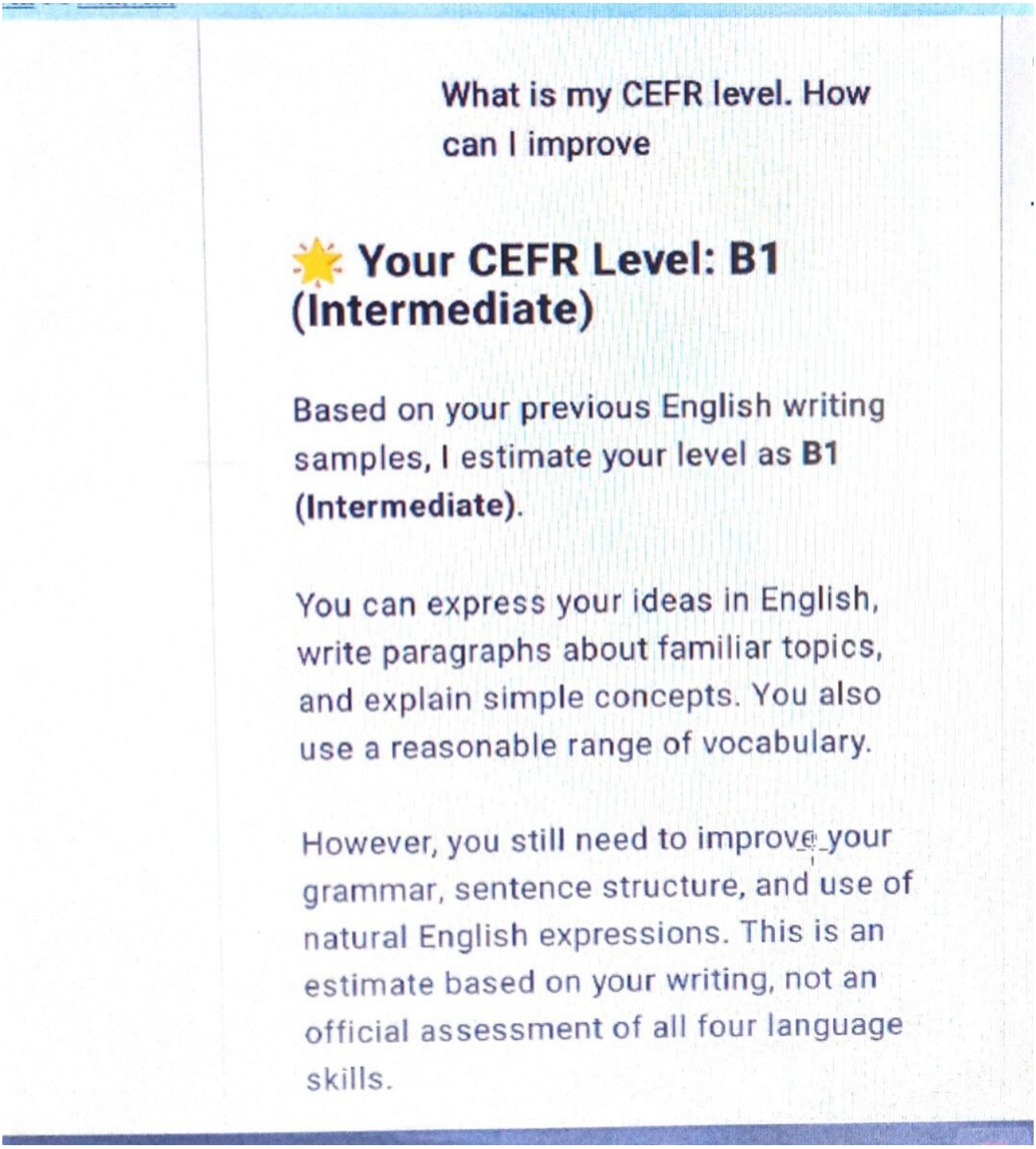
Education

St.Joseph's college of Engineering
B.E.Computer science and Engineering · (2026)

Holy prince matriculation higher secondary school, Madipakkam,Chennai-91
12th · (2025 - 2026)

Holy Prince Matriculation higher secondary school
10th · (2023 - 2024)


Google Chrome isn't your default browser
Set as default
This website uses cookies. For more information, please visit the Privacy notice.
Accept cookies
CAMBRIDGE
Help
Back
Progress
CSE C •
hasinisharli@gmail.com • Last active 14 days ago
Interchange Fifth Edition Level 2
Understanding the data
Digital Workbook
240/240
Completed
236/240
Above target
4/240
Below target
AVG
95%
Average score
10h 29m
Time spent
Terms of use
Privacy notice
Accessibility
Our approach
FAQs
© Cambridge University Press & Assessment 2026
Help
English


What is my CEFR level. How can I improve
Your CEFR Level: B1 (Intermediate)
Based on your previous English writing samples, I estimate your level as B1 (Intermediate).
You can express your ideas in English, write paragraphs about familiar topics, and explain simple concepts. You also use a reasonable range of vocabulary.
However, you still need to improve your grammar, sentence structure, and use of natural English expressions. This is an estimate based on your writing, not an official assessment of all four language skills.

CSE-C
26CS1194

You Choose, We Do It

St. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution)

St. Joseph's Group of Institutions

OMR, CHENNAI - 119

33

EN25C09: Professional English- I (LAB INTEGRATED)

MODEL EXAM I –PRACTICE WORKSHEET

PART - A (15 X 1 = 15 Marks)

Answer ALL questions. Each question carries 1 mark.

1. Fill in the blank with the correct form of the verb given in bracket:

The automated system _____ (monitor) the temperature continuously to ensure safe operation.

Answer: Monitors

2. Fill in the blank with the correct past form of the verb given in bracket:

The engineers _____ (record) the vibration readings during yesterday's trial.

Answer: recorded

3. Complete the sentence with the appropriate comparative form of the adjective in bracket:

The newly developed composite material is significantly _____ (strong) than the conventional material while maintaining a lower weight.

Answer: stronger

4. Fill in the blank with an appropriate modal auxiliary verb: (must/can/might)

Engineers _____ follow safety regulations while operating heavy machinery.

Answer: must

5. Fill in the blank with a suitable modal auxiliary verb expressing capability: (can, may, might)

Modern sensors _____ detect even minor changes in temperature.

Answer: can

6. Combine the sentences into a single coherent sentence using a suitable relative pronoun:

(a) The laboratory developed a new testing device. (b) The device measures material strength with high precision.

Answer: The laboratory developed a new testing device which measures material strength with high precision.

7. Complete the sentence using the correct form of the verb in bracket (First Conditional – If Clause):

If the temperature _____ (rise) above the permissible limit, the cooling system will activate automatically.

Answer: rises

8. Complete the sentence using the correct form of the verb in bracket (Third Conditional – Past Unreal Conditional):

If the technician had inspected the circuit earlier, the equipment _____(not / fail) during operation.

Answer: would not have failed

9. Combine the following sentence using an infinitive of purpose (to + verb):

The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Fill in the blank with the gerund form of the verb in bracket after the preposition:

The research team reduced energy consumption by _____ (optimize) the motor control algorithm.

Answer: optimizing

11. Convert the following active voice into the Impersonal Passive Voice:

The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Answer: _____

12. Fill in the blank with the appropriate adverbial form derived from the adjective in bracket:

The robotic arm moves _____ (smooth) to ensure accurate positioning of the component.

Answer: smoothly

13. Rewrite the direct into indirect speech:

The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Answer: The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.

14. Identify the incorrect noun form and rewrite the sentence correctly:

The laboratory purchased several new equipments for conducting material testing.

Answer: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Fill in the blank using the correct irregular comparative form (better, worse, farther/further, less, more):

Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.

Answer: worse

PART - B (10 X 2 = 20 Marks)

Answer all questions. Each question carries 2 marks.

1. **Rewrite the following active operational and laboratory description in the Simple Past Tense:**

   The engineer calibrates the pressure sensor and verifies its accuracy.

   Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. **Draft two comparative sentences comparing a laptop and a desktop computer in terms of portability and screen size.**

Answer (Portability): A laptop is more portable than a desktop computer.

Answer (Screen Size): A desktop computer generally has a larger screen than a laptop.

3. **Combine the technical statements into a single cohesive sentence using a defining or non-defining relative clause:**

The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Answer: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. **Complete the following conditional sentences logically:**

(a) If the battery voltage drops below the required level,
Answer the system will automatically shutdown.

(b) If the technician had checked the connections earlier,
Answer the equipment would not have malfunctioned.

5. **Convert the following active sentences into the Impersonal Passive Voice suitable for formal engineering reports:**

The laboratory assistant heated the solution to 80°C before testing.

Answer: It was reported that the solution was heated to 80°C before testing.

6. **Rewrite the following direct engineering query into Reported (Indirect) Speech:**

The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Answer: The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.

7. **Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:**

(a) *The engineer received many feedbacks on the new design.*
Answer much feedback

(b) *The students received several useful advices from their teacher.*
Answer some useful advice

8. **Rewrite the sentences using an Infinitive of Purpose or a Gerund construction:**

(a) *The technician wears safety gloves. He wants to protect his hands.* $\rightarrow$
Answer The technician wears safety gloves to protect his hands.

(b) *The cooling fan operates continuously. It prevents the motor from overheating.*
Answer The cooling fan operates continuously to prevent the motor from overheating.

9. **Fill in the blanks with the correct adverbial forms of the bracketed words:**

a) The machine operates efficiently (efficient) to reduce energy consumption.

(b) The technician checked the electrical connections carefully (careful) before starting the machine.

10. **Insert suitable modal auxiliary verbs to match the specified functional meaning:**

(a) The robotic arm can (capability) lift heavy components with precision.

(b) Visitors must not (strict prohibition) enter the testing area without authorization.

PART - C (5 X 13 = 65 Marks)

Answer all questions (Choose either 'a' or 'b' where options are provided). Each question carries 13 marks.

Question 26

- **26. a)** Write a set of eight clear, imperative safety instructions to promote safe and responsible use of the internet and protect users from cyber threats. *(13 Marks)*
  - **Workspace / Planning Area:**
    1.
    2.
    3.
    4.
    5.
    6.
    7.
    8.

Question 27 (Choose either 27.a or 27.b)

27. a) Write a descriptive essay of 250 words about "A Traditional art form in your region and how modern initiatives can preserve its cultural significance." Discuss its historical origins, traditional practices, and ways to promote and sustain it among younger generations. *(13 Marks)*

OR

26. a. 1. Use strong and unique passwords for all your online accounts.

2. Do not share your passwords with anyone.

3. Enable two-factor authentication to protect your accounts.

4. Do not click on suspicious link

5. Avoid sharing personal information with strangers online.

6. Install antivirus software and update it regularly.

7. Do not download files from unknown websites

8. Log out of your accounts when using public computers.

27. b) Write a structured travel plan and descriptive feature (250 words) on "An Educational Trip to a Place of Historical and Cultural Importance " Include travel arrangements, important landmarks, local food experiences, and observations of the region's cultural heritage and traditions. *(13 Marks)*

Workspace / Outline Area:

- *Selected Question:* [27.a / 27.b]
- *Introduction:*
- *Body Paragraph 1:*
- *Body Paragraph 2:*
- *Conclusion:*

Question 28 (Choose either 28.a or 28.b)

28. a) Write a formal Letter of Request to the Librarian of your college seeking permission to extend library working hours during the semester examination period. Explain the academic benefits, the expected student participation, and the arrangements required to ensure proper use of library facilities. *(13 Marks)*

OR

28. b) Draft a formal Letter of Request to the Principal of your college seeking permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution. Mention the objectives of participation, the expected learning outcomes, and the arrangements required. *(13 Marks)*

27. b.

Introduction:

Last month, our college organised an educational trip to madurai, one of the oldest cities in TamilNadu.

Body paragraph 1: Travel Arrangements and Landmarks

-> Travelled by bus
-> First destination - Meenakshi amman temple

Body paragraph 2: Local Food and cultural traditions

-> South Indian dishes
-> Madurai jigarthanda
-> Festivals.
-> cultural life

Conclusion:

The educational trip to Madurai was a memorable experience. It improved our knowledge, architecture and Tamil culture.

28. b. Formal letter to librarian

R. Hasini
First Year Engineering student
Chennai
Date : 20 October 2026

The librarian
[college name]
Chennai.

Subject : Request to Extend Library Working hour during semester Examinations.

Respected sir / madam,

I am writing on behalf of the first year engineering student to request an extention of the library's working hours during the upcoming semester examinations. . . . .

Thank you.

Yours faithfully,

R. Hasini

Workspace / Format Outline:

- *Selected Question:* [28.a / 28.b]
- *Sender's Details / Date:* _____
- *Recipient's Details:* _____
- *Subject:* _____
- *Salutation:* _____
- *Content / Body:* _____
- *Sign-off:* _____

Question 29 (Choose either 29.a or 29.b)

29. a) Compose an analytical essay (250–300 words) comparing the advantages and disadvantages of Artificial Intelligence–Based Assessment versus Traditional Teacher-Based Assessment in engineering education. Evaluate assessment accuracy, feedback time, fairness, and the role of human judgement. *(13 Marks)*

OR

29. b) Write a compare-and-contrast essay of 250–300 words analysing the advantages and disadvantages of Living in Urban Areas versus Rural Areas. Discuss employment opportunities, cost of living, access to education and healthcare, and quality of life. *(13 Marks)*

- **Workspace / Comparative Matrix & Outline:**
  - *Selected Question:* [29.a / 29.b]

Workspace / Format Outline:

- *Selected Question:* [28.a / 28.b]
- *Sender's Details / Date:* _____
- *Recipient's Details:* _____
- *Subject:* _____
- *Salutation:* _____
- *Content / Body:* _____
- *Sign-off:* _____

Question 29 (Choose either 29.a or 29.b)

29. a) Compose an analytical essay (250–300 words) comparing the advantages and disadvantages of Artificial Intelligence–Based Assessment versus Traditional Teacher-Based Assessment in engineering education. Evaluate assessment accuracy, feedback time, fairness, and the role of human judgement. *(13 Marks)*

OR

29. b) Write a compare-and-contrast essay of 250–300 words analysing the advantages and disadvantages of Living in Urban Areas versus Rural Areas. Discuss employment opportunities, cost of living, access to education and healthcare, and quality of life. *(13 Marks)*

- **Workspace / Comparative Matrix & Outline:**
  - *Selected Question:* [29.a / 29.b]

29. b. Living in Urban area vs rural areas.

Criteria 1 vs Criteria 2:

Urban area vs Rural area: Compare employment opportunities, cost of living, education, healthcare, quality of life.

Pros & cons Breakdown:

Urban area: More job opportunities, better education, advanced healthcare...

Rural area: Lower living cost, fresh air, peaceful environment....

Drafting notes:

Introduce urban & rural lifestyle. Compare their advantage & disadvantage. Conclude that both have benefits and challenges.

- *Criteria 1 vs. Criteria 2:*

- *Pros & Cons Breakdown:*

- *Drafting Notes:*

Question 30 (Choose either 30.a or 30.b)

30. a) Draft a personal email to your school friend sharing your experiences during the first semester of your engineering course. Describe your new friendships, classroom activities, adjustment to college life, and expectations for the upcoming semester examinations. *(13 Marks)*

OR

30. b) Draft a personal email to your school friend describing your experience attending the orientation programme for first-year engineering students at your college. Highlight the activities conducted, the introduction to academic facilities and college regulations, and how the programme helped you understand college life and prepare for your academic journey. *(13 Marks)*

- **Workspace / Email Structure:**
  - *Selected Question:* [30.a / 30.b]
  - *To:*
  - *Subject Line:*
  - *Salutation:*
  - *Opening / Check-in:*

30.b.

To : friend@gmail.com

Subject : My college Orientation Programme.

Dear Friend,

How are you? I hope you are doing well. I am excited to share my experience of attending the orientation programme for first year engineering students at my college. It was organised to welcome the new students. several activities were conducted. It helped me to understand how college life differs from school life...

With love,

Hasini

Professional English - I (Lab Integrated)
Model – I – **Practice Worksheet**

Course Code: EN25CO9 | **Subject:** Professional English - I (Lab Integrated)

Exercise 1: Simple Present Tense (Subject-Verb Concord)

Fill in the blanks with the correct form of the verbs given in brackets:

1. Neither the project supervisor nor the lead programmers recognize (recognize / recognizes) the anomaly in the build script.
2. Each of the automated sorting arms requires (require / requires) recalibration after twelve hours of continuous duty.
3. Either the primary sensor or the auxiliary telemetry units transmit (transmit / transmits) packet data to the ground station.
4. The consortium of aerospace contractors agrees (agree / agrees) to adhere to the revised tolerance levels.
5. Physics and material science form (form / forms) the core theoretical foundation for nanoscale fabrication.

Exercise 2: Simple Past Tense (Trial Reporting)

Fill in the blanks with the simple past form of the given verbs:

1. Last month, the metallurgy division conducted (conduct) destructive fatigue testing on the titanium alloy shaft.
2. The field engineers deployed (deploy) thirty IoT water-monitoring nodes across the river basin.
3. During the pressure overload trial, the relief valve opened (open) at 150 psi as designed.
4. The research unit published (publish) their initial benchmark results on low-power RISC-V architectures.
5. In the 2025 dry run, the autonomous navigation algorithm detected (detect) all dynamic road obstacles successfully.

Exercise 3: Comparative Adjectives (Technical Metrics)

Complete each sentence with the appropriate comparative form of the adjective in brackets:

1. Gallium nitride (GaN) semiconductors operate at higher (high) power efficiency than standard silicon transistors.
2. Solid-state batteries deliver a safer (safe) operational threshold under rapid discharge cycles than traditional lithium cells.
3. The revised compiler generates code that executes faster (fast) than the legacy unoptimized routine.
4. Brushless DC motors are considerably more (durable) than brushed equivalents due to reduced mechanical wear.
5. The liquid cooling loop maintained a lower (low) core temperature than the dual fan-sink array.

Exercise 4: Modals of Obligation & Necessity

Fill in each blank with an appropriate modal auxiliary (`must`, `should`, or `have to` / `has to`):

1. Every operator must wear anti-static wrist straps before handling open CMOS motherboards.
2. Hazardous chemical waste must be neutralized prior to disposal into the industrial drain network.
3. Field technicians have to log their entry time into the high-voltage testing bay without fail.
4. To maintain ISO certification, calibration logs must be audited biannually by an external inspector.
5. The test rig must undergo an emergency stop inspection before initiating the high-speed turbine run.

Exercise 5: Modals of Capability & Possibility

Fill in each blank with a suitable modal expressing capability or possibility (`can`, `could`, `may`, `might`):

1. With the upgraded DSP chip, the system can process 4K video streams in real time.
2. If thermal throttling occurs, the processing clock rate might drop by as much as 30%.
3. Advanced laser profilometry can measure surface roughness down to sub-micron tolerances.
4. Unshielded signal lines may pick up electromagnetic interference from nearby induction coils.
5. The modular chassis design can accommodate up to four secondary expansion cards.

Exercise 6: Relative Clauses

Combine each pair of sentences into a single coherent sentence using a suitable relative pronoun (`who`, `which`, `that`, `whose`):

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.
2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.
3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.
4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.
5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

Complete the sentences with the correct form of the verbs in brackets:

1. If the bus voltage exceeds (exceed) 24V, the zener diode will divert the surge to ground.
2. The watchdog timer will reset the CPU if the main thread fails (fail) to ping it within 500 ms.
3. If the operator engages (engage) the safety latch, the hydraulic press will not actuate.
4. If temperature levels rise (rise) above 90°C, the cooling fans will switch to maximum RPM.
5. The system will trigger an auditory alarm if the toxic gas detector registers (register) more than 20 ppm.

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

Fill in the blanks with the past unreal conditional form (`had + past participle` / `would have + past participle`):

1. If the backup generator had started (start) immediately, the data centre servers would not have shut down.
2. The turbine bearing would not have seized if the maintenance crew had lubricated (lubricate) it as scheduled.
3. If the security team had patched the zero-day exploit earlier, the data breach would have been prevented (prevent).
4. The circuit breaker would have tripped (trip) safely if the overcurrent sensor had functioned properly.
5. If the telemetry link had remained (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (`to + verb`):

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.
2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.
3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.
4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.
5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition:

1. The aerodynamic team improved aircraft fuel economy by redesigning (redesign) the winglet profile.
2. Always disconnect the power supply before replacing (replace) defective surface-mount components.
3. The facility was praised for reducing (reduce) volatile organic chemical emissions by 40%.
4. Avoid cleaning (clean) precision optical lenses with abrasive industrial solvents.
5. The digital signal processor isolates target audio signals by filtering (filter) out background ambient noise.

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice:

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.
2. The automation unit tested the robotic arm under a 50 kg dynamic payload.
3. Field engineers calibrated the optical pyrometer prior to ignition.
4. The technicians observed severe cavitation damage along the impeller blades.
5. We store volatile solvents in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverbial form derived from the adjective in brackets:

1. The stepper motor rotates precisely (precise) in 1.8-degree angular increments.
2. The telemetry ground station periodically (periodic) polls all deployed marine buoy sensors.

3. The thermal imaging camera registered the temperature drop rapidly (rapid) during quenching.
4. Structural joints must be welded firmly (firm) to withstand heavy cyclic vibrational stress.
5. The autonomous drone traversed the confined pipeline corridor smoothly (smooth).

Exercise 13: Reported Speech (Interrogative Structures)

Transform each direct technical query into indirect/reported speech:

1. The quality manager asked the technician, "Has the insulation resistance test been documented?"
2. The lead architect enquired, "Can the server cluster handle double the transaction throughput?"
3. The field auditor asked the safety engineer, "Did you inspect the gas leakage relief lines?"
4. The client asked the team, "Will the firmware update fix the intermittent battery drain?"
5. The instructor asked the lab group, "Why did the multimeter show fluctuating voltage readings?"

Exercise 14: Countable and Uncountable Nouns

Identify the incorrect noun form and rewrite each sentence correctly:

1. The procurement officer purchased several new machineries and equipments for the workshop.
2. The laboratory generated a huge amount of test datas during the wind tunnel trials.
3. We need three more softwares licenses to complete the finite element analysis on the model.
4. The field report provided numerous valuable informations regarding the soil composition.
5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

Fill in the blanks using the correct irregular comparative form (`better`, `worse`, `farther` / `further`, `less`, `more`):

1. Because the heat sink had poor thermal paste contact, heat dissipation was worse (bad) than expected.
2. The redesigned microstrip antenna achieved a better (good) signal-to-noise ratio than the baseline patch.
3. The autonomous vehicle traveled 50 km farther (far) on a single charge after the weight reduction overhaul.
4. Switching to asynchronous task queues resulted in less (little) CPU idle time during peak batch processing.

5. We require further (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.

Part B

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

Rewrite the following active operational and lab descriptions into the Simple Past Tense:

1. "The technician inspects the solar arrays and records the peak power conversion value."
2. "The chemical analyzer detects trace mercury levels and sends an automated alert to the environmental dashboard."
3. "Maintenance crews lubricate the heavy conveyor bearings and replace the worn rubber belts before the morning shift."
4. "The supervisory software polls field sensors every five milliseconds and archives temperature trends into a secure database."
5. "The testing team applies a 500-volt dielectric pulse across the stator winding and monitors for insulation breakdown."

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

Draft technical comparative sentences based on the prompts provided:

1. Write two comparative sentences contrasting **Cloud Storage** with **On-Premise Physical Servers** in terms of *initial deployment cost* and *elastic scalability*.
2. Write two comparative sentences evaluating **Fiber-Optic Cables** against **Copper Twisted-Pair Cables** regarding *data transfer bandwidth* and *susceptibility to electromagnetic interference*.
3. Write two comparative sentences contrasting **Electric Vehicles (EVs)** with **Internal Combustion Engine (ICE) Vehicles** focusing on *tailpipe emissions* and *initial purchase price*.
4. Write two comparative sentences comparing **Hydraulic Actuators** with **Electromechanical Actuators** in terms of *raw force output* and *risk of fluid leakage*.
5. Write two comparative sentences comparing **Relational (SQL) Databases** with **Document (NoSQL) Databases** with respect to *schema rigidity* and *horizontal query throughput*.

Exercise 3: Relative Clauses (Sentence Synthesis)

Combine each pair of technical statements into a single cohesive sentence using a defining or non-defining relative clause (`who`, `which`, `that`, `whose`, or `where`):

1. (a) The algorithm analyzes microscopic cracks in turbine blades.

   (b) It was patented by the aeronautics team.

2. (a) The lead chemical engineer authorized the emergency boiler shutdown.

   (b) She noticed an abnormal pressure spike in chamber two.

3. (a) The laboratory houses a 400 kV impulse voltage generator.

   (b) High-voltage insulator puncture tests are conducted there.

4. (a) We engaged an external cybersecurity consultant.

   (b) His firmware vulnerability audit exposed three backdoor ports.

5. (a) The supervisory system utilizes an artificial neural network.

   (b) It predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

Complete the following conditional statements logically and grammatically to depict realistic engineering scenarios:

1. (a) If the operating temperature exceeds 85°C,
   the system will shut down automatically. .

   (b) If the operator had pressed the emergency shutdown switch immediately,
   the safety system would have prevented the accident. .

2. (a) If the supply voltage drops below the 200V threshold,
   the backup power supply will activate. .

   (b) If the network administrator had enabled multi-factor authentication,
   the breach might have been prevented. .

3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel,
   the alarm will sound and workers will evacuate. .

   (b) If the mechanical team had checked the bearing lubrication schedule,
   the bearing would not have failed. .

4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter,
   the AGV will stop immediately. .

   (b) If the thermal sensor had registered the runaway reaction in time,
   the reaction could have been contained. .

5. (a) If the software receives an unauthenticated API token,
   the system will reject the token. .

   (b) If the civil inspectors had reviewed the core sample density tests,
   the inspectors could have detected the weak foundation. .

Exercise 5: Impersonal Passive Voice

Convert the following active sentences into the objective Impersonal Passive Voice suitable for formal engineering reports:

1. "The chemical engineers distilled the crude extract at a constant temperature of 60°C."
2. "The metallurgist polished the cross-section of the weld seam with fine diamond paste."
3. "The quality control division rejected fifteen cast-iron pump casings due to internal porosities."
4. "The technician measured the resistance across the bridge circuit using a calibrated digital ohmmeter."
5. "The flight test crew initiated the emergency aerodynamic braking flaps at an altitude of 3,000 meters."

Exercise 6: Reported Speech (Interrogative Technical Transformations)

Transform the following direct engineering queries into Reported (Indirect) Speech:

1. The safety supervisor asked the intern, "Did you verify the grounding connections on the generator?"
2. The plant director asked the site engineer, "Has the emergency effluent treatment plant received clearance?"
3. The IT security lead enquired of the developer, "Why did you disable the firewall rule on the database server?"
4. The client asked the project manager, "Can the bridge structure withstand Category 4 hurricane winds?"
5. The research guide asked the postgraduate scholar, "How many sample trials did you conduct under cryogenic conditions?"

Exercise 7: Countable vs. Uncountable Nouns

Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:

1. (a) We need to install several new software and machineries in the CAD laboratory.

   (b) The team collected plenty of useful informations from the field test.

2. (a) The technician ordered ten new apparatuses and modern electronic equipments for the antenna lab.

(b) High-speed train operations require advanced transport infrastructures and dedicated electrical powers.

3. (a) The environmental sensor gathers extensive atmospheric datas and pollutions indices every minute.

   (b) The construction crew unloaded tons of sands, cements, and steel debris at the worksite.

4. (a) All electrical hardwares and wirings must comply with local safety legislations.

   (b) The data scientist discovered several interesting knowledges after running the machine learning clustering.

5. (a) Laboratory staffs must clear away all glasswares and acidic wastes before leaving the premises.

   (b) The project failed due to a lack of proper fundings and technical researches.

Exercise 8: Infinitives of Purpose & Gerund Constructions

*Rewrite the following pairs of sentences using either an **Infinitive of Purpose (`to + verb`)** or a **Gerund construction after a preposition (`by / for + -ing`)**:*

1. (a) A pressure relief valve is installed. It vents excess boiler steam.

   (b) The automated script runs every midnight. It archives legacy server logs.

2. (a) Automotive engineers installed an intercooler. It reduces the compressed air intake temperature.

   (b) The technicians analyzed the structural stress points. They placed piezoelectric strain gauges along the girder.

3. (a) High-performance heatsinks feature copper heatpipes. They conduct heat away from the silicon die.

   (b) The factory cut water consumption by 30%. They implemented a closed-loop recycling filtration system.

4. (a) Safety guards are mounted over rotating flywheel belts. They prevent accidental worker contact.

   (b) The algorithm optimizes memory bandwidth. It pre-fetches sequential array indices into L2 cache.

5. (a) The electronics lab installed an uninterruptible power supply (UPS). It protects servers against sudden voltage sags.

   (b) The metallurgy group enhanced surface hardness. They diffused nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision

Fill in the blanks with the appropriate adverbial forms derived from the words provided in brackets:

1. (a) The drone adjusted its flight path automatically (automatic) to avoid overhead power cables.

   (b) The software patch was integrated flawlessly (flawless) into the production server.

2. (a) The CNC cutting head traced the complex automotive profile accurately (accurate).

   (b) The turbine governor responded instantly (instant) to the unexpected grid frequency drop.

3. (a) The hazardous gas extraction vents operate continuously (continuous) throughout the electroplating process.

   (b) Technicians handled the radioactive isotope capsule cautiously (cautious) using robotic arm manipulators.

4. (a) The optical pyrometer recorded the furnace temperature consistently (consistent) across three production shifts.

   (b) The structural frame buckled catastrophically (catastrophic) under the simulated earthquake test load.

5. (a) The firmware routine checks for incoming serial packets periodically (periodic) every hundred microseconds.

   (b) The titanium alloy workpiece was machined smoothly (smooth) to a mirror finish.

Exercise 10: Modals of Capability, Obligation, and Prohibition

Complete the sentences by inserting an appropriate modal auxiliary verb that conveys the specific operational meaning specified in brackets:

1. (a) The microcontroller $_{can}$ (capability) drive up to six stepper motors simultaneously.

   (b) Laboratory staff $_{must\ not}$ (strict prohibition) handle corrosive reagents without full face shields.

2. (a) Before entering the cleanroom chamber, all personnel $_{must}$ (mandatory obligation) put on antistatic footwear.

   (b) Given the high ambient humidity, water droplet condensation $_{may}$ (moderate possibility) form inside the optical housing.

3. (a) Unauthorized maintenance personnel $_{must\ not}$ (strict prohibition) override the high-voltage interlock switch.

   (b) The upgraded wireless transmitter $_{could}$ (past capability) broadcast telemetry signals up to forty kilometers.

4. (a) Quality inspection teams $_{should}$ (strong recommendation / advice) recalibrate measuring calipers every ninety days.

   (b) The digital ammeter is reading zero amperes; the auxiliary fuse $_{must}$ (logical deduction / certainty) be blown.

5. (a) Industrial robots $_{can}$ (capability) perform repetitive spot welds with sub-millimeter precision.

   (b) Visitors $_{must\ not}$ (strict prohibition) enter the laser testing enclosure while the amber warning beacon is active.

Topic 1: Writing Imperative Safety Instructions (CO1 | BL3)

Target: 8 clear, numbered, imperative instructions using modals/action verbs.

1. **Question 1 :** Write a set of eight clear, imperative safety instructions to be strictly displayed and followed inside a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.
2. **Question 2 :** Write a set of eight clear, imperative safety instructions to be strictly displayed and observed inside a High-Voltage Electrical Machines & Drives Laboratory to prevent electric shock, arc flash hazards, and accidental short circuits during motor loading experiments.
3. **Question 3:** Write a set of eight clear, imperative safety guidelines to be posted in an Advanced Manufacturing and CNC Machining Workshop to prevent injuries from high-speed spindle operations, hot metallic chip expulsion, coolant spills, and moving cutter heads.

Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4)

Target: Descriptive essay of 250 words integrating community customs, festive meals, and sustainable practices.

1. **Question 1 :** Write a descriptive essay of 250 words about "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core Spirit." Discuss community customs, traditional festive meals, and sustainable celebration methods.
2. **Question 2 :** Write a descriptive essay of 250 words on "A Traditional Temple Car or Float Festival in Your Hometown and Sustainable Measures to Minimize Single-Use Plastics and Noise Pollution." Detail communal rituals, traditional food offerings, and eco-friendly crowd management methods.
3. **Question 3 :** Write a descriptive essay of 250 words on "Celebrating Regional Folk Art and Handicraft Fairs Using Renewable Solar Micro-Grids and Zero-Waste Composting." Discuss the cultural atmosphere, heritage delicacies, and modern environmental protection strategies.

Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4)

Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

1. **Question 1 :** Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam." Include route details, logistics, local cultural food experiences, and observations of the plant's operational landscape.
2. **Question 2 :** Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to an Integrated Solar and Wind Hybrid Energy Farm." Detail route planning, transit logistics, local culinary specialties sampled en route, and operational observations of the wind turbine park and photovoltaic arrays.
3. **Question 3 :** Write a structured travel plan and descriptive feature (250 words) on "A Technical Field Visit to an Automated Seawater Reverse Osmosis (RO) Desalination Plant." Describe transportation details, regional coastal food experiences, and observations of intake pumping stations, membrane filtration racks, and brine discharge outlets.

Topic 4: Formal Letters of Request (CO3 | BL4)

Target: Formal business letter layout with sender/receiver address, clear subject line, technical rationale, specific requirements, and courteous sign-off.

1. **Question 1 :** Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon. Outline project objectives and required software resources.

2. **Question 2 (Original - Industrial Visit):** Write a formal Letter of Request to the Plant Manager of an Industrial Robotics & Automation Factory seeking approval for an industrial plant visit for 50 first-year engineering students, outlining the intended learning outcomes.
3. **Question 3 (Practice A - Workshop Facilities):** Draft a formal Letter of Request to the Head of the Mechanical Engineering Department seeking permission and machine shop access after college hours for your student team to fabricate and test an energy-efficient electric go-kart for an inter-collegiate design challenge.
4. **Question 4 (Practice B - Sponsorship & Duty Leave):** Draft a formal Letter of Request to the College Principal / Dean of Student Affairs requesting financial sponsorship and duty leave for a five-member multidisciplinary student team selected to present a drone surveillance prototype at a National Defense Innovation Expo.

Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5)

Target: Analytical essay (250–300 words) using comparative transitions (whereas, conversely, in contrast, on the other hand) to evaluate technical trade-offs.

1. **Question 1 :** Write a comprehensive Compare and Contrast essay (250–300 words) analyzing the Positives and Negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion Engine (ICE) Vehicles. Address battery life, manufacturing footprint, operating costs, and charging network feasibility.
2. **Question 2 :** Compose an analytical essay comparing the Positives and Negatives of Remote / Virtual Internships versus On-Site Physical Internships for early-stage engineering students, evaluating hands-on skill acquisition, schedule flexibility, mentorship, and professional networking.
3. **Question 3 :** Write an analytical Compare and Contrast essay (250–300 words) evaluating the Positives and Negatives of Cloud Computing Infrastructure versus On-Premise Physical Data Centers for engineering enterprises. Address upfront capital expenditure, data security compliance, recurring overheads, and workload scalability.
4. **Question 4 :** Write a balanced Compare and Contrast essay (250–300 words) discussing the Positives and Negatives of 3D Printing (Additive Manufacturing) versus Traditional CNC Milling (Subtractive Manufacturing) in prototype fabrication. Evaluate material wastage, production speed for batch runs, surface finish accuracy, and geometric complexity limits.

Topic 6: Personal Email Writing (CO5 | BL6)

Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic transitions, and structured study plans.

1. **Question 1 :** Draft a Personal Email to your high school mentor or teacher sharing your first-semester experiences at engineering college. Highlight the transition to independent campus life, your initial laboratory projects, and how you are preparing for your upcoming end-semester examinations.

2. **Question 2 :** Draft a personal email to your high school science teacher reflecting on your first term in engineering. Focus on adapting to hands-on engineering lab coursework, participating in your first technical club or hackathon, and your strategy for the upcoming university theory and practical exams.
3. **Question 3 :** Draft a personal email to your elder sibling or college alumnus seeking advice on balancing academic requirements with co-curricular technical clubs. Describe your daily schedule, challenges in managing lab record submissions, and ask for their tips on semester exam revision.

Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6)

Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical activities and safety measures.

1. **Question 1 :** Write an engaging, well-organized Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch Station, detailing transport logistics, budget calculations, planned technical visits, and safety protocols.
2. **Question 2 :** Write a comprehensive Travel Plan and Budgetary Itinerary (250 words) for a 4-day student industrial exposure tour to a Major Commercial Port and Maritime Dry-Dock Facility. Include transport arrangements, per-student cost estimates (food, travel, stay), daily technical briefing stops, and industrial dockyard safety protocols.
3. **Question 3 :** Write an engaging Travel Plan and Budgetary Itinerary (250 words) for a 4-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Center. Detail rail transit bookings, category-wise expenditure estimates, technical shop-floor visits, and site visitor safety rules.

ANSWERS FOR PART C:

QUESTION :3(WRITING IMPERATIVE SAFETY INSTRUCTIONS)

1. Wear safety glasses, safety shoes and close-fitting clothes; tie back long hair and remove rings, watches and loose jewellery.

2. Do not wear gloves near a rotating spindle or cutter, because they can get caught and pull your hand in.

3. Close the safety door and check that all guards and interlocks are in place before you start the machine; never bypass them.

4. Clamp the workpiece and tighten the cutting tool firmly, and remove the chuck key or wrench before you start the spindle.

5. Keep your hands away from the spindle and cutter head, and stop the machine completely before you measure, clean or change a tool.

6. Clear hot metal chips only with a brush or chip hook once the machine has stopped; never use your bare hands or compressed air.

7. Wipe up coolant and oil spills immediately and keep the floor dry and free of chips to prevent slips and falls.

8. Run the program in dry-run or single-block mode first, keep your hand near the emergency stop button, and never leave a running machine unattended.

QUESTION : 2(FREE WRITING)

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated every January to thank the Sun God, the soil and the farm animals for a good harvest. For four days, the whole community comes together, and the festival shows how closely our culture is connected with agriculture.The celebration begins with Bhogi, when families clean their houses and throw away old things.

On Thai Pongal, the main day, fresh rice is cooked with milk and

jaggery in a new clay pot in the open courtyard. When the milk boils over, everyone shouts "Pongalo Pongal!" On Mattu Pongal, farmers decorate and worship their cattle, and on Kaanum Pongal families visit relatives. Neighbours draw colourful kolams in front of their houses, and villages organise games and cultural programmes.

QUESTION:1(FORMAL LETTER WRITING)

From:

The Project Team,

Department of Computer Science and Engineering,

St. Joseph's College of Engineering, Chennai.

To:

The Head of the Department,

Department of Computer Science and Engineering,

St. Joseph's College of Engineering, Chennai.

Date: 03 November 2026

Subject: Request for Permission and Laboratory Access After College Hours for National AI Hackathon Preparation

Respected Sir/Madam,

We are writing to request your permission to use the Computer Science Laboratory after regular college hours to prepare for an upcoming National AI Hackathon.

Our project team aims to develop an innovative AI-based solution to address a real-world problem. The main objectives of our project are to identify a practical problem, design an effective AI model, develop a user-friendly application, test its performance and present a working prototype at the hackathon. This opportunity will help us improve our programming, teamwork, problem-solving and technical skills.

For successful project development, we require access to computers with a stable internet connection and the following software resources:

Python programming language.

Visual Studio Code or PyCharm.

Jupyter Notebook.

Libraries such as NumPy, Pandas and TensorFlow.

Git and GitHub for version control.

Necessary web development tools.

We assure you that we will maintain discipline, handle laboratory equipment responsibly and follow all college rules. We will also ensure that the laboratory is kept clean and all systems are properly shut down after use.

We kindly request you to grant us permission to access the laboratory after college hours until the hackathon preparation is completed.

Thank you for your time and consideration.

Yours faithfully,

The Project Team

Department of Computer Science and Engineering

St. Joseph's College of Engineering

QUESTION:1(COMPARE AND CONTRAST)

Electric Vehicles vs Conventional Internal Combustion Engine Vehicles

The automobile industry is undergoing a major transformation with the introduction of Electric Vehicles (EVs). Electric vehicles and conventional Internal Combustion Engine (ICE) vehicles are two major transportation options. Both have advantages and

disadvantages in terms of cost, environmental impact, maintenance and convenience.

Electric Vehicles (EVs) operate using electricity stored in rechargeable batteries, whereas ICE vehicles run on petrol or diesel. One major advantage of EVs is that they produce no exhaust emissions while driving, helping to reduce air pollution. In contrast, ICE vehicles release harmful gases and contribute to environmental pollution. However, EV battery manufacturing requires minerals such as lithium and cobalt, and their extraction and processing have an environmental impact. Battery disposal and recycling are also important concerns.

Another important difference is battery life and maintenance. EV batteries gradually lose their capacity over time and may require replacement after several years, which can be expensive. On the other hand, ICE vehicles do not depend on large rechargeable batteries but require regular engine servicing, oil changes and fuel-system maintenance.

Regarding operating costs, EVs are generally cheaper to operate because electricity is often less expensive than petrol or diesel. They also have fewer moving parts, reducing maintenance expenses. However, their initial purchase price can be higher. ICE vehicles are widely available but involve continuous fuel expenses and regular maintenance costs.

Charging infrastructure is another challenge for EVs. Charging stations are still limited in some areas, and charging takes longer than refuelling conventional vehicles. ICE vehicles offer greater convenience because petrol and diesel stations are widely available.

In conclusion, EVs offer environmental and economic benefits in the long term, while ICE vehicles provide convenience and established infrastructure. Both technologies have limitations, but EVs represent a promising step towards sustainable transportation.

QUESTION :1(EMAIL WRITING)

Dear Sir/Madam,

I hope you are doing well. I am writing this email to share my experiences during my first semester at engineering college.

My transition from school life to college life has been an exciting and memorable experience. Initially, I found it a little challenging to adjust to the new environment, independent learning and college schedule. However, I gradually became comfortable with the campus, made new friends and started participating in various academic activities. College life has taught me to manage my time, take responsibility and become more independent.

During this semester, I have gained practical experience through laboratory sessions, especially in C programming and computer fundamentals. I have started writing simple programs, solving logical problems and understanding how theoretical concepts are applied practically. These activities have improved my technical knowledge and problem-solving skills.

At present, I am preparing for my upcoming end-semester examinations. I have created a study timetable, revising important concepts, practising programming questions and solving previous question papers. I am also focusing on improving my understanding of difficult topics and managing my time effectively.

I am grateful for the knowledge, guidance and encouragement you provided during my school days. The values and discipline I learned from you continue to help me in my college journey.

I hope to achieve good results and make you proud.

Thank you for always inspiring me.

Yours sincerely,

[Your Name]

QUESTION :3(TRAVEL PLANNING AND BUDGETARY ITINERARIES)

Travel Plan and Budgetary Itinerary for a 4-Day Technical Study

Tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Centre

A four-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Centre will provide engineering students with valuable practical knowledge about modern railway technology, manufacturing and automation.

Day 1 – Journey and Arrival: Students will travel by train from Chennai to the selected destination using pre-booked reserved tickets. After arrival, they will check into pre-booked accommodation. An introductory session will be conducted on railway engineering and electric locomotive technology.

Day 2 – Locomotive Manufacturing Visit: Students will visit the manufacturing facility to observe locomotive assembly, electric motors, braking systems, power transmission and quality testing. Engineers will explain the manufacturing process and modern technologies used in railway production.

Day 3 – Automated Signalling Centre: Students will visit an approved railway signalling centre to understand automatic signalling, electronic interlocking, train monitoring systems and railway safety operations. They will also attend technical presentations by railway experts.

Day 4 – Return Journey: Students will participate in a discussion session, share their learning experiences and return to Chennai by train.

Estimated Budget per Student:

Category Amount

Train tickets (round trip) ₹3800

Accommodation (3 nights) ₹1,500

Food expenses ₹900

Local transportation ₹400

Entry and permission charges ₹200

Miscellaneous expenses ₹200

Total ₹4,000

Safety Rules: Students must wear safety helmets, shoes and protective equipment wherever required. They must follow staff instructions, avoid restricted areas, maintain discipline and never touch operating machinery or electrical equipment without permission.

This technical tour will enhance students' practical knowledge, encourage innovation and provide valuable exposure to modern railway engineering and automation systems.

